

ASSIGNMENT NO. 2

ReverseServer.java

```
import ReverseModule.*;

import org.omg.CosNaming.*;

import org.omg.CosNaming.NamingContextPackage.*;

import org.omg.CORBA.*;

import org.omg.PortableServer.*;

class ReverseServer

{

    public static void main(String args[])

    {

        try

        {

            ORB orb = ORB.init(args, null);

            POA rootpoa = POAHelper.narrow(orb.resolve_initial_references("RootPOA"));

            rootpoa.the_POAManager().activate();

            Reverselmpl reverselmpl = new Reverselmpl();

            org.omg.CORBA.Object ref = rootpoa.servant_to_reference(reverselmpl);

            Reverse href = ReverseHelper.narrow(ref);

            org.omg.CORBA.Object objRef = orb.resolve_initial_references("NameService");

            NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef);

            String name = "Reverse";

            NameComponent path[] = ncRef.to_name(name);

            ncRef.rebind(path, href);
```

```

        System.out.println("Reverse Server ready and waiting...");

        orb.run();
    }
    catch(Exception e)
    {
        e.printStackTrace();
    }
}

```

ReverseImpl.java

```

import ReverseModule.ReversePOA;

class ReverseImpl extends ReversePOA
{
    ReverseImpl()
    {
        super();
        System.out.println("Reverse Object Created");
    }

    public String reverse_string(String name)
    {
        StringBuffer str = new StringBuffer(name);
        str.reverse();
        return ("Server Send " + str);
    }
}

```

ReverseClient.java

```

import ReverseModule.*;
import org.omg.CosNaming.*;

```

```
import org.omg.CosNaming.NamingContextPackage.*;

import org.omg.CORBA.*;

import java.io.*;

class ReverseClient
{
    public static void main(String args[])
    {
        try
        {
            ORB orb = ORB.init(args, null);

            org.omg.CORBA.Object objRef = orb.resolve_initial_references("NameService");
            NamingContextExt ncRef = NamingContextExtHelper.narrow(objRef);

            String name = "Reverse";
            Reverse reverseImpl = ReverseHelper.narrow(ncRef.resolve_str(name));

            BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
            System.out.println("Enter String:");
            String str = br.readLine();

            String result = reverseImpl.reverse_string(str);
            System.out.println(result);
        }
        catch(Exception e)
        {
            e.printStackTrace();
        }
    }
}
```

Output

Terminal 1:

```
C:\Users\anuja\OneDrive\Documents>cd DS_prac
C:\Users\anuja\OneDrive\Documents\DS_prac>idlj -fall ReverseModule.idl
C:\Users\anuja\OneDrive\Documents\DS_prac>javac *.java ReverseModule/*.java
```

Terminal 2:

```
C:\Users\anuja\OneDrive\Documents>cd DS_prac
C:\Users\anuja\OneDrive\Documents\DS_prac>orbd -ORBInitialPort 1050
```

Terminal 3 :

```
C:\Users\anuja\OneDrive\Documents>cd DS_prac

C:\Users\anuja\OneDrive\Documents\DS_prac>java ReverseServer -ORBInitialPort 1050 -
ORBInitialHost localhost

Reverse Object Created

Reverse Server ready and waiting...
```

Terminal 4:

```
C:\Users\anuja\OneDrive\Documents>cd DS_prac

C:\Users\anuja\OneDrive\Documents\DS_prac>java ReverseClient -ORBInitialPort 1050 -
ORBInitialHost localhost

Enter String:

Anuja

Server Send ajunA

C:\Users\anuja\OneDrive\Documents\DS_prac>java ReverseClient -ORBInitialPort 1050 -
ORBInitialHost localhost

Enter String:

Welcome to Distributed Systems

Server Send smetsyS detubirtsID ot emocleW
```